

Infinite Sequences and Series

In this chapter, we ~~can~~ study the topic of Infinite Series. Infinite series give us precise ways to express many numbers and functions as arithmetic sums with infinitely many terms.

For example, we will learn that

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$$

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\text{and } \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

We need to develop a method to make sense of such expressions.

Everyone knows how to add two numbers together, or even several. But how do you add together infinitely many numbers? Or, how do you add together infinitely many powers of x ? In this chapter we

answer these questions. Limits play a major role in the development of infinite series.

The terms "sequence" and "series" both are distinct. A sequence is a type of infinite list, whereas a series is an infinite sum. To understand the infinite sums described by series, we need to first study infinite sequences.

Representing Sequences

A sequence is a list of numbers

$$a_1, a_2, a_3, \dots, a_n, \dots$$

in a given order. Each of a_1, a_2, a_3 and so on represents a number. These are the terms of the sequence.

For example, the sequence

$$1^2, 2^2, 3^2, \dots, n^2, \dots$$

has first term $a_1 = 1^2$, second term $a_2 = 2^2$, and n th term $a_n = n^2$. The integer n is called the index of a_n and indicates where a_n occurs in the list.

Note that the sequence $1^2, 2^2, 3^2, 4^2, \dots$ is not the same as the sequence $2^2, 3^2, 1^2, 4^2, \dots$. Order is important.

We can think of the sequence

$$a_1, a_2, a_3, \dots, a_n, \dots$$

as a function that sends 1 to a_1 , 2 to a_2 , 3 to a_3 , and in general sends the positive integer n to the n th term a_n . More precisely, an infinite sequence of numbers is a function whose domain is the set of positive integers.

Sequences can be described by writing rules that specify their terms, such as

$$a_n = \sqrt{n}, \quad b_n = n^3 - 1, \quad c_n = (-1)^n n^2, \quad d_n = (-1)^{n+1} \frac{1}{n}, \\ e_n = (-1)^{n+1}$$

or by listing terms:

$$\{a_n\} = \{\sqrt{1}, \sqrt{2}, \sqrt{3}, \dots, \sqrt{n}, \dots\}$$

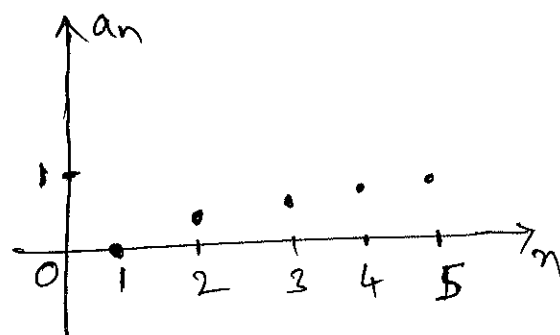
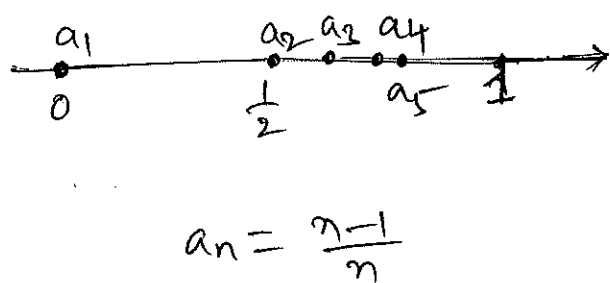
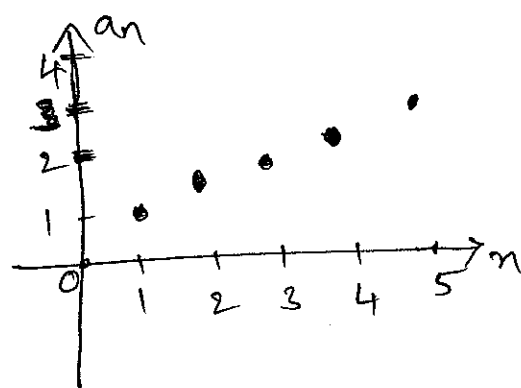
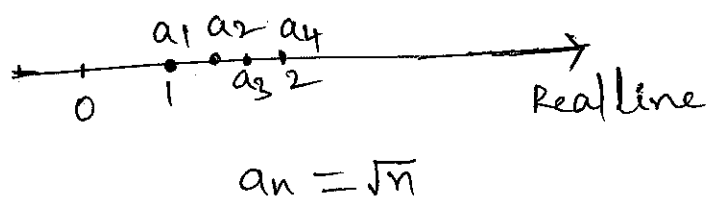
$$\{b_n\} = \{0, 7, 26, 63, 124, \dots, n^3 - 1, \dots\}$$

$$\{c_n\} = \{-1, 4, -9, 16, -25, \dots, (-1)^n n^2, \dots\}$$

$$\{d_n\} = \{1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots, (-1)^{n+1} \frac{1}{n}, \dots\}$$

$$\{e_n\} = \{1, -1, 1, -1, 1, -1, \dots, (-1)^{n+1}, \dots\}$$

Sequences can be represented graphically in two ways. The first one, sequences can be represented as points on the real line, and the second one is sequences can be represented as points on the plane where the horizontal axis n is the index number of the term and the vertical axis a_n is its value.



Sometimes the numbers in a sequence approach a single value as the index n increases.

This happens in the sequence

$$\left\{ 1, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \dots, \frac{1}{n^2}, \dots \right\}$$

②

whose terms approach 0 as n gets large, and in the sequence

$$\left\{ \frac{2}{1}, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \dots, \frac{n+1}{n}, \dots \right\}$$

whose terms approach 1 as n gets large.

On the other hand, sequences like

$$\{1^3, 2^3, 3^3, 4^3, \dots, n^3, \dots\}$$

have ~~the~~ terms that get larger than any number as n increases, and the sequences like

$$\{1, -1, +1, -1, \dots, (-1)^{n+1}, \dots\}$$

bounce back and forth between 1 and -1, never converging to a single value.

Convergence and Divergence

Definition:- The sequence $\{a_n\}$ converges to the number L if for every positive number ϵ there corresponds an integer N such that $|a_n - L| < \epsilon$ whenever $n > N$.

If no such number L exists, we say that $\{a_n\}$ diverges.

If $\{a_n\}$ converges to L , we write $\lim_{n \rightarrow \infty} a_n = L$ or

simply $a_n \rightarrow L$, and call L the limit of the sequence.

① By definition, show that

i) $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

ii) $\lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$

iii) $\lim_{n \rightarrow \infty} \frac{1}{3^n} = 0$

iv) $\lim_{n \rightarrow \infty} [\sqrt{n+1} - \sqrt{n}] = 0$

Solution

i) Let $\varepsilon > 0$ be given. We must show that there exists an integer N such that $|\frac{1}{n} - 0| < \varepsilon$ whenever $n > N$.

$$\text{Since } |\frac{1}{n} - 0| < \varepsilon \Rightarrow |\frac{1}{n}| < \varepsilon \Rightarrow n > \frac{1}{\varepsilon}.$$

If N is any integer greater than $\frac{1}{\varepsilon}$, the implication will hold for all $n > N$. This proves that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

ii) Let $\varepsilon > 0$ be given and $a_n = \frac{1}{n^2}$

$$\text{Now } |a_n - 0| = |\frac{1}{n^2} - 0| = |\frac{1}{n^2}| = \frac{1}{n^2} < \varepsilon$$

$$\Rightarrow n^2 > \frac{1}{\varepsilon} \Rightarrow n > \frac{1}{\sqrt{\varepsilon}}$$

If N is a positive integer $> \frac{1}{\sqrt{\varepsilon}}$, then $|a_n - 0| < \varepsilon$ $\forall$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$$

$n > N$

(iii) Let $\varepsilon > 0$ be given and $a_n = \frac{1}{3^n}$

(4)

$$\text{Now } |a_n - 0| = \left| \frac{1}{3^n} - 0 \right| = \left| \frac{1}{3^n} \right| = \frac{1}{3^n} < \varepsilon$$

$$\Rightarrow 3^n > \frac{1}{\varepsilon}$$

$$\Rightarrow n \ln 3 > \ln\left(\frac{1}{\varepsilon}\right)$$

$$\Rightarrow n > \frac{\ln\left(\frac{1}{\varepsilon}\right)}{\ln 3}$$

If N is a positive integer $> \frac{\ln\left(\frac{1}{\varepsilon}\right)}{\ln 3}$, then

$$|a_n - 0| < \varepsilon \quad \forall n > N$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{3^n} = 0$$

i.e., the sequence $\left\{ \frac{1}{3^n} \right\}$ converges to 0.

(iv) Let $\varepsilon > 0$ be given and let $a_n = \sqrt{n+1} - \sqrt{n}$

$$\text{Now } |\sqrt{n+1} - \sqrt{n}| = \left| \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} \times \sqrt{n+1} + \sqrt{n} \right|$$

$$= \left| \frac{1}{\sqrt{n+1} + \sqrt{n}} \right| < \frac{1}{\sqrt{n} + \sqrt{n}} = \frac{1}{2\sqrt{n}} < \frac{1}{\sqrt{n}} < \varepsilon$$

$$\Rightarrow \sqrt{n} > \frac{1}{\varepsilon} \Rightarrow n > \frac{1}{\varepsilon^2}$$

If N is a positive integer $> \frac{1}{\varepsilon^2}$, then $|a_n - 0| < \varepsilon \quad \forall n > N$

$$\text{i.e. } \lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$$

Definition:- the sequence $\{a_n\}$ diverges to infinity if for every number M there is an integer N such that for all n larger than N , $a_n > M$. If this condition holds we write $\lim_{n \rightarrow \infty} a_n = \infty$ (or) $a_n \rightarrow \infty$

Similarly, if for every number m there is an integer N such that for all $n > N$, we have $a_n < m$, then we say $\{a_n\}$ diverges to negative infinity and we write $\lim_{n \rightarrow \infty} a_n = -\infty$ (or) $a_n \rightarrow -\infty$.

Example ② show that

- i) the sequence $\{n\}$ diverges to $+\infty$.
- ii) The sequence $\{-n^2\}$ diverges to $-\infty$.
- iii) The sequence $\{n^2\}$ diverges to ∞ .
- iv) The sequence $\{\log \frac{1}{n}\}$ diverges to $-\infty$.
- v) The sequence $\{2^n\}$ diverges to ∞ .

A sequence may diverge without diverging to infinity or negative infinity. Examples of such divergence sequences are

$$\{1, -1, 1, -1, \dots, (-1)^{n+1}, \dots\}$$

$$\{1, -2, 3, -4, 5, -6, \dots, (-1)^{n+1}n, \dots\}$$

$$\{1, 0, 2, 0, 3, 0, \dots, \dots\}$$

$$\{1, \frac{1}{2}, 2, \frac{1}{3}, 3, \frac{1}{4}, \dots\}$$

Example ②: If $a_n = 1 + \frac{(-1)^n}{2n}$, find the least positive integer N such that

$$|a_n - 1| < \frac{1}{10^3} \quad \forall n > N.$$

ii) If $a_n = 2 + \frac{(-1)^n}{n^2}$, find the least positive integer N such that

$$|a_n - 2| < \frac{1}{10^4} \quad \forall n > N.$$

Soln:

$$|a_n - 1| = \left| \frac{(-1)^n}{2n} \right| = \frac{1}{2n}$$

$$|a_n - 1| < \frac{1}{10^3} \Rightarrow \frac{1}{2n} < \frac{1}{10^3} \Rightarrow 2n > 1000$$

$$\Rightarrow n > 500$$

Taking $N = 500$, we have $|a_n - 1| < \frac{1}{10^3} \quad \forall n > N.$

ii) please try yourself.

Theorem ①:- Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers, and let A and B be real numbers. The following rules hold if $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$.

i) Sum Rule: $\lim_{n \rightarrow \infty} (a_n + b_n) = A + B$

ii) Difference Rule: $\lim_{n \rightarrow \infty} (a_n - b_n) = A - B$

iii) Constant Multiple Rule: $\lim_{n \rightarrow \infty} (k \cdot b_n) = k B$

iv) Product Rule: $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = A \cdot B$

v) Quotient Rule: $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}$ if $B \neq 0$.

Theorem ② The sandwich theorem for sequences

Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences of real numbers

If $a_n \leq b_n \leq c_n$ holds for all n beyond some index N ,

and if $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$

also.

Note:- If $|b_n| \leq c_n$ and $c_n \rightarrow 0$, then $b_n \rightarrow 0$ because

$$-c_n \leq b_n \leq c_n$$

Example i) $\frac{\sin n}{n} \rightarrow 0$

(ii) $\frac{\cos^2 n}{2^n} \rightarrow 0$

(6)

(iii) $\frac{(-1)^n}{n} \rightarrow 0$

(iv) If $|a_n| \rightarrow 0$ then $a_n \rightarrow 0$ ($\because -|a_n| \leq a_n \leq |a_n|$)

Theorem (3) the continuous function theorem for sequences

Let $\{a_n\}$ be a sequence of real numbers. If $a_n \rightarrow L$ and if f is a function that is continuous at L and defined at all a_n , then $f(a_n) \rightarrow f(L)$.

Example:- show that $\{2^{1/n}\}$ converges to 1.

Sol:- the sequence $\{\frac{1}{n}\}$ converges to 0.

Taking $f(x) = 2^x$ and $L = 0$

We see that $2^{1/n} = f(\frac{1}{n}) \rightarrow f(L) = 2^0 = 1$.

The sequence $\{2^{1/n}\}$ converges to 1.

Using L'Hôpital's Rule:-

Theorem (4) Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence of real numbers such that $a_n = \cancel{f(n)} f(n)$ for $n \geq n_0$.

then $\lim_{n \rightarrow \infty} a_n = L$ whenever $\lim_{x \rightarrow \infty} f(x) = L$.

Note: the above theorem connection between

$\lim_{n \rightarrow \infty} a_n$ and $\lim_{x \rightarrow \infty} f(x)$. It enables us to use L'Hôpital's rule to find the limit of some sequences.

Example 1, show that $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$

ii) show that $\lim_{n \rightarrow \infty} \frac{n}{2^n} = 0$

Sol: i) The function $\frac{\ln x}{x}$ is defined for all $x \geq 1$ and agrees with the given sequence at positive integers.

Therefore, $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{x \rightarrow \infty} \frac{\ln x}{x}$ by theorem (4).

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0 \text{ by L'Hôpital's Rule}$$

We conclude that $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$

$$\text{ii) } \lim_{n \rightarrow \infty} \frac{n}{2^n} = \lim_{n \rightarrow \infty} \frac{1}{2^n \ln 2}$$

$$= \frac{1}{\ln 2} \lim_{n \rightarrow \infty} \frac{1}{2^n} = \frac{1}{\ln 2} \cdot 0 = 0$$

Example:- Does the sequence whose n th term is

$$a_n = \left(1 + \frac{7}{n}\right)^n \text{ converge? If so, find}$$

$$\lim_{n \rightarrow \infty} a_n.$$

Sol:- The limit leads to the indeterminate form 1^∞ .

We can apply L'Hopital's Rule if we first change the form to $\infty \cdot 0$ by taking the natural logarithm of a_n :

$$\ln a_n = \ln \left(1 + \frac{7}{n}\right)^n = n \ln \left(1 + \frac{7}{n}\right)$$

$$\text{Then, } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} n \ln \left(1 + \frac{7}{n}\right) \quad \infty \cdot 0 \text{ form}$$

$$= \lim_{n \rightarrow \infty} \frac{\ln \left(1 + \frac{7}{n}\right)}{\frac{1}{n}} \quad \frac{0}{0} \text{ form}$$

$$= \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{1 + \frac{7}{n}}\right) \cdot \left(-\frac{7}{n^2}\right)}{\left(-\frac{1}{n^2}\right)}$$

$$= \lim_{n \rightarrow \infty} 7 \cdot \left(\frac{n}{n+7}\right) = 7$$

Since $\ln a_n \rightarrow 7$ and $f(x) = e^x$ is continuous, Theorem (4) tells us that $a_n = f(a_n) = e^{\ln a_n} \rightarrow e^7$.
The sequence $\{a_n\}$ converges to e^7 .

Theorem ⑤ The following six sequences converge to the limits listed below:

$$\text{i), } \lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$$

$$\text{ii), } \lim_{n \rightarrow \infty} n^{1/n} = 1$$

$$\text{iii), } \lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$$

$$\text{iv), } \lim_{n \rightarrow \infty} x^n = 0 \quad (|x| < 1)$$

$$\text{v), } \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x \quad (\text{any } x)$$

$$\text{vi), } \lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad (\text{any } x)$$

In formulas iii) through vi), x remains fixed as $n \rightarrow \infty$.

Theorem ⑥ If $\{a_n\}$ is a sequence such that $a_n \neq 0$ for any n and $\frac{a_{n+1}}{a_n} \rightarrow l$ or $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = l$ where $|l| < 1$, then $a_n \rightarrow 0$ or $\lim_{n \rightarrow \infty} a_n = 0$.

Example ① Show that $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$

Sol: $a_n = \frac{n!}{n^n} \quad a_{n+1} = \frac{(n+1)!}{(n+1)^{n+1}}$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!}$$

$$= \frac{n^n}{(n+1)^n} = \frac{1}{\left(1 + \frac{1}{n}\right)^n}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e} < 1$$

Hence, $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$.

Example 2 Show that $\lim_{n \rightarrow \infty} \frac{100^n}{n!} = 0$

Sol:- $a_n = \frac{100^n}{n!}$ $a_{n+1} = \frac{100^{n+1}}{(n+1)!}$

$$\frac{a_{n+1}}{a_n} = \frac{100^{n+1}}{(n+1)!} \times \frac{n!}{100^n} = \frac{100}{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{100}{n+1} = 0 < 1$$

Hence, $\lim_{n \rightarrow \infty} \frac{100^n}{n!} = 0$

i.e., the sequence $\left\{ \frac{100^n}{n!} \right\}$ converges to 0.

Example 3 Show that $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$

Sol:- please try yourself

Theorem ⑦ Cauchy's first theorem on limits

Statement:- If $\{a_n\}$ converges to l , then the sequence

$\{x_n\}$, where $x_n = \frac{a_1 + a_2 + \dots + a_n}{n}$ also converges to l .
(OR)

$$\lim_{n \rightarrow \infty} a_n = l \Rightarrow \lim_{n \rightarrow \infty} \frac{a_1 + a_2 + \dots + a_n}{n} = l$$

Example ① show that $\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1}}{n} = 0$

sol:- Let $a_n = \frac{1}{2n-1}$, then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{2n-1} = 0$

By Cauchy's first theorem on limits

$$\lim_{n \rightarrow \infty} \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1}}{n} = 0$$

② show that $\lim_{n \rightarrow \infty} \frac{1 + 2^{1/2} + 3^{1/3} + \dots + n^{1/n}}{n} = 1$

③ show that $\lim_{n \rightarrow \infty} \frac{\left(\frac{2}{1} + \frac{3}{2} + \frac{4}{3} + \dots + \frac{n+1}{n}\right)}{n} = 1$

④ show that $\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2}}{n} = 0$

Theorem ⑧ Cauchy's second theorem on limits. ⑨

Statement:- If $\{a_n\}$ is a sequence of positive terms, then $\lim_{n \rightarrow \infty} (a_n)^{1/n} = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$ provided the limit on the right hand side exists, whether finite or infinite.

Example:- ① Show that $\lim_{n \rightarrow \infty} n^{1/n} = 1$

Sol:- Let $a_n = n$, then $\frac{a_{n+1}}{a_n} = \frac{n+1}{n} = 1 + \frac{1}{n}$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} 1 + \frac{1}{n} = 1$$

Hence, $\lim_{n \rightarrow \infty} (a_n)^{1/n} = \lim_{n \rightarrow \infty} n^{1/n} = 1$, by Theorem ⑧

② find $\lim_{n \rightarrow \infty} (n!)^{1/n}$

Sol:- Let $a_n = n!$, then $\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{n!} = n+1$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} n+1 = \infty$$

By Cauchy's second theorem on limits,

$$\lim_{n \rightarrow \infty} (a_n)^{1/n} = \lim_{n \rightarrow \infty} (n!)^{1/n} = \infty.$$

③ prove that $\lim_{n \rightarrow \infty} \left(\frac{n^n}{n!} \right)^{1/n} = e.$

Sol: Let $a_n = \frac{n^n}{n!}$, then $a_{n+1} = \frac{(n+1)^{n+1}}{(n+1)!}$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^{n+1}}{(n+1)!} \times \frac{n!}{n^n} = \left(\frac{n+1}{n}\right)^n = \left(1 + \frac{1}{n}\right)^n$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

By Cauchy's second theorem on limits,

$$\lim_{n \rightarrow \infty} (a_n)^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{n^n}{n!}\right)^{1/n} = e$$

④ Find $\lim_{n \rightarrow \infty} \left\{ \frac{(3n)!}{(n!)^3} \right\}^{1/n}$

Sol: Let $a_n = \frac{(3n)!}{(n!)^3}$, $a_{n+1} = \frac{3(n+1)!}{((n+1)!)^3}$

$$\frac{a_{n+1}}{a_n} = \frac{(3n+1)(3n+2)(3n+3)}{(n+1)^3}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 27$$

By Cauchy's second theorem on limits, $\lim_{n \rightarrow \infty} (a_n)^{1/n} = 27$.

⑤ Evaluate $\lim_{n \rightarrow \infty} \left[\frac{(n+1)(n+2) \dots (n+n)}{n^n} \right]^{1/n}$

please try yourself.

Exercise: show that

$$\text{i), } \lim_{n \rightarrow \infty} (n^2 + n)^{1/n} = 1$$

$$\text{ii), } \lim_{n \rightarrow \infty} \frac{\sin \frac{n\pi}{3}}{\sqrt{n}} = 0$$

$$\text{iii), } \lim_{n \rightarrow \infty} \frac{\cos \frac{n\pi}{4}}{n^2} = 0$$

Recursive definitions: so far, we have calculated each a_n directly from the value of n . But sequences are often defined ~~by~~ recursively by giving

- i) the value(s) of the initial term or terms, and
- ii) A rule, called a recursion formula, for calculating any later term from terms that precede it.

Example i) The statements $a_1 = 1$ and $a_n = a_{n-1} + 1$ for $n > 1$ define the sequence $1, 2, 3, \dots, n, \dots$ of positive integers.

With $a_1 = 1$, we have $a_2 = a_1 + 1 = 2$, $a_3 = a_2 + 1 = 3$, and so on.

ii) The statements $a_1 = 1$, $a_2 = 1$, and $a_{n+1} = a_n + a_{n-1}$ for $n > 2$ define the sequence $1, 1, 2, 3, 5, \dots$, of Fibonacci numbers. With $a_1 = 1$ and $a_2 = 1$, we have $a_3 = 1 + 1 = 2$, $a_4 = 2 + 1 = 3$, $a_5 = 3 + 2 = 5$, and so on.

Bounded Monotonic Sequences:-

Two concepts that play a key role in determining the convergence of a sequence are those of a bounded sequence and a monotonic sequence.

① Bounded Sequence:-

(a) A sequence $\{a_n\}$ is bounded from above if there exists a number M such that $a_n \leq M \forall n$. The number M is an upper bound for $\{a_n\}$. If M is an upper bound for $\{a_n\}$ but no number less than M is an upper bound for $\{a_n\}$, then M is the least upper bound for $\{a_n\}$.

(b) A sequence $\{a_n\}$ is bounded from below if there exists a number m such that $a_n \geq m \forall n$. The number m is a lower bound for $\{a_n\}$. If m is a lower bound for $\{a_n\}$ but no number greater than m is a lower bound for $\{a_n\}$, then m is the greatest lower bound for $\{a_n\}$.

(c) If $\{a_n\}$ is bounded from above and below, then $\{a_n\}$ is bounded. If $\{a_n\}$ is not bounded, then we say that $\{a_n\}$ is unbounded sequence.

Example:-

i) the sequence $1, 2, 3, 4, \dots, n, \dots$ has no upper bound. However, it is bounded below. The number 1 is the greatest lower bound of the sequence.

ii) The sequence $-1, -2, -3, -4, \dots, -n, \dots$ has no lower bound. However, it is bounded above. The number -1 is the least upper bound of the sequence.

iii) The sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots, \frac{n}{n+1}, \dots$ is bounded. The sequence bounded above by 1 and below by $\frac{1}{2}$. The number 1 is the least upper bound and $\frac{1}{2}$ is the greatest lower bound of the sequence.

② Monotonic Sequences:-

(a) A sequence $\{a_n\}$ is nondecreasing if $a_n \leq a_{n+1}$ for all n . That is $a_1 \leq a_2 \leq a_3 \leq \dots$

(b) the sequence $\{a_n\}$ is nonincreasing if $a_n \geq a_{n+1}$ for all n .

(c) The sequence $\{a_n\}$ is monotonic if it is either nondecreasing or nonincreasing.

Example:

- i) The sequence $1, 2, 3, \dots, n, \dots$ is nondecreasing.
- ii) The sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots, \frac{n}{n+1}, \dots$ is nondecreasing.
- iii) The sequence $\frac{2}{1}, \frac{3}{2}, \frac{4}{3}, \dots, \frac{n+1}{n}, \dots$ is nonincreasing.
- iv) The sequence $1, \frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \dots, \frac{1}{2^n}, \dots$ is nonincreasing.
-ng.
- v) The constant sequence $3, 3, 3, \dots, 3, \dots$ is both nondecreasing and nonincreasing.
- vi) The sequence $1, -1, 1, -1, \dots, (-1)^{n+1}, \dots$ is not monotonic.

Theorem: The Monotonic Sequence theorem.

If a sequence $\{a_n\}$ is both bounded and monotonic, then the sequence converges.

Note that the above theorem does not say that convergent sequences are monotonic.

For example, the sequence $\left\{ \frac{(-1)^{n+1}}{n} \right\}$ converges and is bounded, but it is not monotonic.

Theorem:- Every convergent sequence is bounded. ⁽¹²⁾

Note ①: The converse of the above is not true i.e., a bounded sequence is not necessarily convergent.

For example, the sequence $\{(-1)^n\}$ which is bounded but it is not convergent.

Note ②: Boundedness is a necessary condition for the convergence of a sequence. If a sequence is not bounded, it cannot be convergent.

For example, the sequence $\{n^2\}$ is not convergent because it is unbounded.

Theorem: i) A nondecreasing sequence that is bounded from above always converges to its least upper bound.

ii) A nonincreasing sequence that is bounded from below always converges to its greatest lower bound.

Theorem: i) A nondecreasing sequence which is not bounded above diverges to ∞ .

ii) A nonincreasing sequence which is not bounded below diverges to $-\infty$.

Theorem: Every monotonic sequence either converges or diverges
(OR)

A monotonic sequence is never oscillatory.

Theorem: Every convergent sequence has a unique limit.
(OR)

A sequence cannot converge to more than one limit.

Theorem: If the sequence $\{a_n\}$ converges to l , then the sequence $\{|a_n|\}$ converges to $|l|$.

Note: The converse of this theorem is not true.

For example, if $a_n = (-1)^n$, then $|a_n| = |(-1)^n| = 1$

which is constant sequence and hence $\{|a_n|\}$ converges to 1, but the sequence $\{a_n\}$ is not convergent.

Theorem: The sequence $\{a_n\}$ converges to 0 $\Leftrightarrow$ the sequence of absolute values $\{|a_n|\}$ converges to 0.

Example: prove that the sequences, whose n th terms are given below, are monotonic.

i) $\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^n}$

ii) $\frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1}$

iii) $\frac{3n+7}{4n+8}$

iv) $a_1 = 1$ and $a_n = \sqrt{2 + a_{n-1}}$, $n \geq 2$

Sol:-

$$a_n = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^n}$$

$$a_{n+1} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^n} + \frac{1}{2^{n+1}}$$

$$a_{n+1} - a_n = \frac{1}{2^{n+1}} > 0 \quad \forall n$$

$$\Rightarrow a_{n+1} > a_n \quad \forall n$$

$\therefore \{a_n\}$ is monotonically increasing sequence.

ii) $a_n = \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1}$

$$a_{n+1} = \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n-1} + \frac{1}{2n} + \frac{1}{2n+1}$$

$$\therefore a_{n+1} - a_n = \frac{1}{2n} + \frac{1}{2n+1} - \frac{1}{n}$$

$$= \frac{-1}{2n(2n+1)} < 0$$

$$\Rightarrow a_{n+1} < a_n \quad \forall n$$

$\therefore \{a_n\}$ is monotonically decreasing

iii) $a_n = \frac{3n+7}{4n+8}$, Then $a_{n+1} = \frac{3n+10}{4n+12}$

$$a_{n+1} - a_n = \frac{3n+10}{4n+12} - \frac{3n+7}{4n+8}$$

$$= \frac{-1}{4(n+3)(n+2)} < 0 \quad \forall n$$

$$\Rightarrow a_{n+1} < a_n \quad \forall n$$

$\therefore \{a_n\}$ is monotonically decreasing sequence.

iv) $a_1 = 1$, $a_2 = \sqrt{2+a_1} = \sqrt{2+1} = \sqrt{3} > 1 = a_1$

$$\Rightarrow a_2 > a_1$$

Now suppose $a_n > a_{n-1}$ for some $n \in \mathbb{N}$, $n > 1$

$$\text{Then } 2 + a_n > 2 + a_{n-1}$$

$$\Rightarrow \sqrt{2+a_n} > \sqrt{2+a_{n-1}}$$

$$\Rightarrow a_{n+1} > a_n$$

$\therefore$ By mathematical induction, $a_{n+1} > a_n \quad \forall n$.

$\Rightarrow \{a_n\}$ is monotonically increasing.

Example: Discuss the convergence of sequence $\{a_n\}$ where

i) $a_n = \frac{n+1}{n}$

ii) $a_n = \frac{n}{n^2+1}$

iii) $a_n = 1 + \frac{1}{2} + \frac{1}{3^2} + \dots + \frac{1}{3^n}$

Sol: i, $a_n = \frac{n+1}{n} = 1 + \frac{1}{n}$

$$a_{n+1} = 1 + \frac{1}{n+1}$$

$$a_{n+1} - a_n = \frac{1}{n+1} - \frac{1}{n} = -\frac{1}{n(n+1)} < 0$$

$$\Rightarrow a_{n+1} < a_n \quad \forall n$$

$\Rightarrow \{a_n\}$ is monotonically decreasing sequence.

Also, $a_n = 1 + \frac{1}{n} > 1 \quad \forall n$

$\Rightarrow \{a_n\}$ is bounded from below.

Since $\{a_n\}$ is monotonically decreasing and bounded from below, it is convergent.

ii, please try yourself.

iii, please try yourself.

Example:- prove that the sequence $\left\{ \frac{2n-7}{3n+2} \right\}$

i, monotonically increasing ii, is bounded and

iii, tends to the limit $\frac{2}{3}$.

Sol:- Let $a_n = \frac{2n-7}{3n+2}$

i)

$$a_{n+1} = \frac{2n-5}{3n+5}$$

$$a_{n+1} - a_n = \frac{2n-5}{3n+5} - \frac{2n-7}{3n+2}$$

$$= \frac{25}{(3n+5)(3n+2)} > 0 \quad \forall n$$

$$\Rightarrow a_{n+1} > a_n \quad \forall n$$

$\Rightarrow \{a_n\}$ is monotonically increasing.

ii)

The sequence $\left\{ \frac{2n-7}{3n+2} \right\} = \left\{ -1, -\frac{3}{8}, -\frac{1}{11}, \frac{1}{14}, \dots \right\}$

clearly $a_n > -1 \quad \forall n$

$$\text{Also, } 1 - a_n = 1 - \frac{2n-7}{3n+2} = \frac{n+9}{3n+2} > 0 \quad \forall n$$

$$\Rightarrow 1 - a_n > 0 \quad \forall n$$

$$\Rightarrow 1 > a_n \quad \forall n$$

$$\text{Thus, } -1 \leq a_n < 1 \quad \forall n$$

$\Rightarrow$ The sequence $\{a_n\}$ is bounded.

iii) Since $\{a_n\}$ is monotonically increasing and bounded above, it is convergent.

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{2n-7}{3n+2} = \lim_{n \rightarrow \infty} \frac{2 - \frac{7}{n}}{3 + \frac{2}{n}} = \frac{2}{3}$$

$\therefore \{a_n\}$ converges to $\frac{2}{3}$.

Ex 1, Prove that the sequence whose n th term is $\frac{3n+4}{2n+1}$

- i) is monotonically decreasing
- ii) is bounded and
- iii) converges to $\frac{1}{2}$

Sol, Please try yourself.

Ex 1 prove that the sequence $\{a_n\}$ defined by

$$a_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!}$$

is convergent and that $2 \leq \lim_{n \rightarrow \infty} a_n \leq 3$.

Example 1 Show that the sequence $\{a_n\}$ defined by $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2a_n}$ converges to 2

Sol, $a_1 = \sqrt{2}$, $a_2 = \sqrt{2a_1} = \sqrt{2\sqrt{2}}$

$$\because \sqrt{2} > 1 \Rightarrow 2\sqrt{2} > 2$$

$$\Rightarrow \sqrt{2\sqrt{2}} > \sqrt{2}$$

$$\Rightarrow a_2 > a_1$$

Suppose $a_n > a_{n-1}$, then $2a_n > 2a_{n-1}$

$$\Rightarrow \sqrt{2a_n} > \sqrt{2a_{n-1}}$$

$$\Rightarrow a_{n+1} > a_n \quad \forall n$$

By mathematical induction, $a_{n+1} > a_n \forall n$

$\therefore \{a_n\}$ is monotonically increasing sequence

Now $a_1 = \sqrt{2} < 2$

Suppose $a_n < 2$, then $2a_n < 4$

$$\Rightarrow \sqrt{2a_n} < 2$$

$$\Rightarrow a_{n+1} < 2$$

$\therefore$ By mathematical induction, $a_n < 2 \forall n$

$\therefore \{a_n\}$ is bounded above.

Since $\{a_n\}$ is monotonically increasing and bounded above, it is convergent.

Let $\lim_{n \rightarrow \infty} a_n = l$ then $\lim_{n \rightarrow \infty} a_{n+1} = l$

$$\therefore a_{n+1} = \sqrt{2a_n}$$

$$\Rightarrow a_{n+1}^2 = 2a_n$$

$$\text{As } n \rightarrow \infty, \quad l^2 = 2l$$

$$\Rightarrow l^2 - 2l = 0 \Rightarrow l(l-2) = 0$$

$$\Rightarrow l = 0 \text{ or } l = 2$$

But $l \neq 0 \because a_1 = \sqrt{2}$ and it is monotonically increasing
 $\therefore l = 2$

Ex 1 - Show that the sequence $\{a_n\}$, where $a_1 = 1$ and $a_n = \sqrt{2 + a_{n-1}}$, $\forall n \geq 2$ is convergent and converges to 2.

Sol:- please try yourself.

Ex 2 - Prove that the sequence $\{a_n\}$ defined by $a_1 = \sqrt{7}$, $a_{n+1} = \sqrt{7 + a_n}$ converges to the positive root of the equation $x^2 - x - 7 = 0$.

Sol:- please try yourself.

Example:- If a_1, b_1 are two positive unequal numbers and a_n, b_n are defined as $a_n = \frac{1}{2}(a_{n-1} + b_{n-1})$,

$$b_n = \sqrt{a_{n-1} b_{n-1}} \text{ for } n \geq 2$$

prove that two sequences $\{a_n\}$ and $\{b_n\}$ are monotonic, one increasing and the other decreasing and that they tend to the same limit.

Sol:- Let $a_1 > b_1$

for any two positive numbers, $A \cdot M > G \cdot M$

$$\therefore a_n > b_n \forall n.$$

$$\text{Also } a_{n+1} = \frac{1}{2}(a_n + b_n) < \frac{1}{2}(a_n + a_n) = a_n$$

$\therefore \{a_n\}$ is monotonic decreasing

$$\Rightarrow a_1 > a_2 > a_3 > a_4 > \dots$$

Again $b_{n+1} = \sqrt{a_n \cdot b_n} > \sqrt{b_n \cdot b_n} = b_n$

$$b_{n+1} > b_n \quad \forall n$$

$\therefore \{b_n\}$ is monotonic increasing

$$\text{i.e., } b_1 < b_2 < b_3 < b_4 < \dots$$

Now

$$a_n = \frac{1}{2}(a_{n-1} + b_{n-1}) > \frac{1}{2}(b_{n-1} + b_{n-1}) = b_{n-1}$$

$$a_n > b_{n-1} > b_{n-2} > \dots > b_2 > b_1$$

$$\Rightarrow a_n > b_1 \quad \forall n$$

$\therefore$ This implies that the sequence $\{a_n\}$ is bounded below and being monotonic decreasing is convergent.

~~Again~~ Again, $b_n = \sqrt{a_{n-1} \cdot b_{n-1}} < \sqrt{a_{n-1} \cdot a_{n-1}} = a_{n-1}$

$$\text{i.e., } b_n < a_{n-1}$$

$$\Rightarrow b_n < a_{n-1} < a_{n-2} < \dots < a_2 < a_1$$

$$\Rightarrow b_n < a_1 \quad \forall n$$

$\therefore \{b_n\}$ is bounded above and being monotonic increasing is convergent.

Now let $\lim_{n \rightarrow \infty} a_n = l$ and $\lim_{n \rightarrow \infty} b_n = m$

$$\text{Since } a_n = \frac{1}{2}(a_{n-1} + b_{n-1})$$

$$\Rightarrow 2a_n = a_{n-1} + b_{n-1}$$

$$\therefore 2a_{n+1} = a_n + b_n$$

$$\Rightarrow \lim_{n \rightarrow \infty} (2a_{n+1}) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

$$\Rightarrow 2l = l + m$$

$$\Rightarrow l = m$$

$\therefore \{a_n\}$ and $\{b_n\}$ converge to the same limit.

Ex:- If $a_1 > 0, b_1 > 0$ and $a_n = \sqrt{a_{n-1} b_{n-1}}$ and $b_n = \frac{2a_{n-1}b_{n-1}}{a_{n-1} + b_{n-1}}$

prove that i) $\{a_n\}$ and $\{b_n\}$ are monotonic, the one increasing and the other decreasing, ii) $\{a_n\}$ and $\{b_n\}$ both converge to a common limit.

Sol:- Please try yourself.

Ex:- prove that the sequence $\{a_n\}$ where

$$a_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \text{ is convergent and}$$

its limit lies between $\frac{1}{2}$ and 1.

Subsequence: If the terms of one sequence appear in another sequence in their given order, we call the first sequence a subsequence of the second.

For example, the sequences $\{2, 4, 6, 8, \dots\}$ and $\{1, 3, 5, 7, \dots\}$ are subsequences of the sequence $\{1, 2, 3, 4, 5, 6, 7, 8, \dots\}$.

Theorem: If a sequence $\{a_n\}$ converges to L , then every subsequence of $\{a_n\}$ also converges to L .

Note: If all subsequences of a sequence $\{a_n\}$ converge to the same limit L , only then $\{a_n\}$ converges to L .

Theorem: If the subsequences $\{a_{2n}\}$ and $\{a_{2n+1}\}$ of a sequence $\{a_n\}$ converge to the same limit L , then the sequence $\{a_n\}$ converges to L .

(OR)

For a sequence $\{a_n\}$ the terms of even index are denoted by a_{2k} and the terms of odd index by a_{2k+1} ; prove that if $a_{2k} \rightarrow L$ and $a_{2k+1} \rightarrow L$, then $a_n \rightarrow L$.

Note: ~~Prove that~~ Two subsequences of a sequence $\{a_n\}$ have different limits $L_1 \neq L_2$, then $\{a_n\}$ diverges.

For example, $\{(-1)^n\}$ diverges.